

Rudra Patel

U.S. Citizen | 224-607-9378 | rudrap26176@gmail.com | [linkedin.com/in/rudrapatel09](https://www.linkedin.com/in/rudrapatel09) | rudras-website.vercel.app

Education

Lewis University

May 2026

B.S. Computer Science, Concentration in Artificial Intelligence - GPA 3.9, Dean's List 6x

Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Big Data Systems (Apache Spark, Hadoop, Kafka, Hive)

University of Illinois at Urbana-Champaign (UIUC)

MCS Computer Science - Admitted Fall 2026 - AI/ML for Enterprise Applications

Experience

Software Engineer

May 2025 – May 2026

Hometown Liquor - E Commerce Platform

- Increased monthly revenue by **45%** to **\$25k** by building an e-commerce platform from scratch with **150+ users** and **770+** products using **React**, **Java Spring Boot**, and **PostgreSQL**
- Secured REST API traffic for **92+** endpoints with Spring Security by implementing stateless JWT auth, role-based admin access, request rate limiting, and audit logging across all REST API endpoints
- Built a hybrid AI product search feature using **PostgreSQL** full-text search and AI-generated embeddings for vector similarity search, returning natural language results in under **1s** on average across **770+ products**
- Reduced manual inventory checks by **90%** by automating POS inventory sync with a third-party API, batch processing catalog updates, updating stock quantities, and logging inventory changes
- Improved deployment reliability with **Docker** containerization and CI/CD, and wrote **85+** unit and integration tests covering auth, product, inventory, and API service behavior

Research Assistant

April 2026 - June 2026

Lewis University

- Built a dataset management platform for Lewis Research Lab to centralize, document, and search across all datasets using **React**, and **Go**
- Implemented an AI-powered search engine using **Go**, combining **PostgreSQL** full-text search and semantic vector search, improving search accuracy by **40%** across all lab datasets
- Designed chunked concurrent dataset uploads and constant-memory streaming downloads in Go, handling dataset files up to **50 GB** without timeouts or server memory pressure
- Engineered an upload-time AI pipeline that automatically generates labels, class distributions, and summaries for every dataset within **3–5 seconds**, using a bounded file profiler to validate and clean LLM output before storing metadata

Projects

Distributed LLM Inference Gateway | Go, PostgreSQL, Redis, React, TypeScript, Docker

- Building an open-source, cost-aware LLM gateway in **Go** that standardizes OpenAI, Anthropic, and local Ollama models behind a unified REST API with per-request tracing and structured logging
- Implemented exact-match Redis cache, load tested to handle **9,000+** requests/s, reducing p99 response latency from **2.1s** to **15ms**
- Building an async **PostgreSQL** database pipeline to track AI usage, latency, and costs, with an AI-powered caching layer that detects similar user queries to drastically reduce expensive API calls

AI Recipe Service | TypeScript, Java, Spring Boot, AWS, PostgreSQL, Docker, Terraform

[Link](#)

- Built a **Java** Spring Boot backend with **12** REST API endpoints across auth, recipe generation, history, exports, metrics, and admin operations, secured with stateless JWT auth, BCrypt hashing, and user-scoped access control
- Added fault-tolerant generation workflows with rate limiting, retry logic, and circuit breaking, then validated reliability with **139** total backend unit and integration tests covering auth, validation, malformed model responses, audit persistence, and S3 exports
- Defined **AWS** infrastructure in **Terraform** for ECS Fargate, ALB, RDS, S3, CloudWatch, IAM, private networking, and encrypted storage

Real-Time Event Processor | Node.js, Express.js, TypeScript, Redis, TimescaleDB, Docker

[Link](#)

- Built a distributed event pipeline similar to Kafka processing **2.4M+** events (**28 events/sec**) with zero message loss using **Node.js**, **Express.js**, and **Redis Streams** (consumer groups)
- Reduced dashboard query latency by **55x (1.828s to 33ms)** using TimescaleDB, eliminating live computation at query time
- Cut database transactions by **60–70%** via batch processing (**100 events/batch**) with schema validation, and conflict-safe inserts
- Built a REST API using **Express.js** and **Redis** caching with p95 response times of **5-38ms** and 5-10s TTL, serving **2.4M+** records

Hackathons & Competitions

ASA DataFest - American Statistical Association | Python, Pandas, Matplotlib, Seaborn

April 2026

- Built an ETL pipeline in a 48-hour competition using **Python**, cleaning and joining **7.6M+** patient records from global healthcare dataset
- Analyzed **79K** patient records using Python classification algorithms, identifying an **83.6%** hypertension and **74.4%** diabetes disengagement rate, and presented findings and outreach recommendations to competition judges within 48 hours

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, Go, HTML, CSS

Frameworks & Tools: Spring Boot, React, Express.js, Node.js, REST API, Docker, GitHub Actions, Linux, Bash Scripting

Database & Cloud: PostgreSQL, Redis, Elasticsearch, MySQL, Firebase, AWS (ECS, RDS, S3, CloudWatch, IAM), GCP

Practices & Big Data: AI-Assisted Development, REST APIs, Agile, Microservices, Apache Spark, Hadoop, Kafka, Hive